

7 (a) Define *magnetic flux density*.

[1]

Magnetic flux density is defined as the force acting per unit length per unit current on a conductor carrying current placed at right angles to the magnetic field.

- (b) A 1.0 m long wire carrying a current of 2.0 A is placed in a magnetic field. When it is aligned with XX' as shown in Fig. 7.1A, it experiences a force of 0.50 N that is directed into the page. When it is aligned with YY' as shown in Fig. 7.1B, it experiences a force of 0.80 N that is directed out of the page.

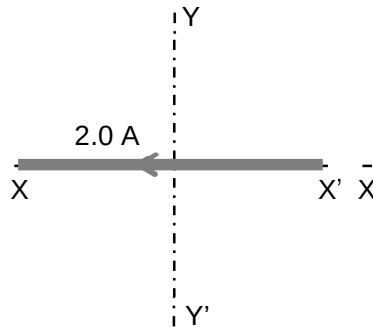


Fig. 7.1A

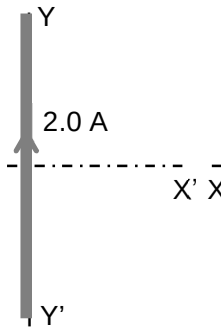


Fig. 7.1B

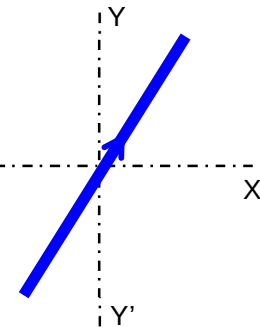


Fig. 7.1C

- (i) Calculate the magnetic flux density of the field present. [3]

$$\begin{aligned}(F/L)_x &= B_y I \rightarrow B_y = 0.50 / 2.0 = 0.25 \text{ T } \uparrow \\ (F/L)_y &= B_x I \rightarrow B_x = 0.80 / 2.0 = 0.40 \text{ T } \leftarrow \\ B &= \sqrt{(0.25^2 + 0.40^2)} = 0.47 \text{ T}\end{aligned}$$

- (ii) Determine the position of the wire such that it experiences a force of the maximum magnitude. Draw accurately the position of the wire in Fig. 7.1C. [2]

$$\begin{aligned}\theta &= \tan^{-1} (0.25 / 0.40) = 32^\circ \\ \text{Direction of B is } \square 32^\circ &\rightarrow \text{Alignment of wire for maximum force is } \square 58^\circ\end{aligned}$$

- (c) The current-carrying wire in (b) has a magnetic field of its own. Show, by calculation, that it is impossible to find a point near the wire such that its magnetic flux density at that point has the same magnitude as the field present. [2]

$$\begin{aligned}B &= \mu_0 I / 2\pi d \rightarrow 0.47 = 2 \times 10^{-7} \times 2.0 / d \\ d &= 8.5 \times 10^{-7} \text{ m} \\ \text{distance is much smaller than thickness of wire} &\rightarrow \text{not possible}\end{aligned}$$